Mini-Lesson 3: Impulse–Momentum Theorem

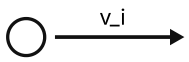
AP Physics 1 · Unit 4 · Do with the mini-lesson before class Mon 11/16. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

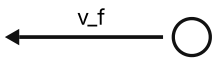
Name _____ Period _____

A Impulse is a change in momentum

BEFORE



AFTER the bat



J = Δp = _____ − _____

Δ means _____. Units: _____ = _____

p goes from +3 to +8 kg·m/s: J = _____

B From F = ma to FΔt = Δp

F = ma, and a = _____, so F = m _____

Multiply both sides by Δt: _____

J = _____ × _____ = _____

Impulse points the same direction as the _____.

Newton's 2nd law in momentum form: F_{net} = _____

C Same Δp, different force

Stopping a 0.50 kg ball at 10 m/s: Δp = 5.0 kg·m/s every time

Δt = 0.01 s (hard floor)

Δt = 0.2 s (catching into a pillow)

F = Δp ÷ Δt: 0.01 s → F = _____ 0.2 s → F = _____

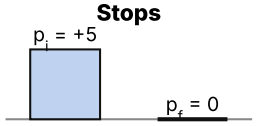
For the same Δp, a longer Δt gives a _____ force.

Doubling Δt makes F _____.

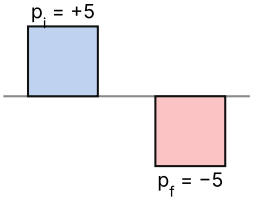
Examples: _____

D Stopping vs. bouncing (toward wall is +)

Stops



Bounces back



Stops: Δp = _____ Bounces: Δp = _____ kg·m/s

A bounce changes p _____ because the object must stop and _____.

E Worked example: 0.150 kg baseball, pitched at 30.0 m/s, hit straight back at 50.0 m/s, contact 0.001 s

+ direction	p_i	p_f	J = Δp	F_{avg} = J / Δt
toward the batter	_____	_____	_____	_____

F Check questions: show your work

1. 0.60 kg basketball from rest, 70 N for 0.20 s. Final speed?

work:

v_f = _____

2. 0.20 kg ball hits a wall at 6.0 m/s, bounces back at 4.0 m/s. Δp? (toward wall +)

work:

Δp = _____

3. Crash time tripled, same Δp. Average force becomes _____ because _____

Big idea in one sentence:

_____

_____